

Easy Gym Life Repository Model Tree

Architecture, file dependencies, runtime flow, platform adapters, and tracked-file inventory

Generated from the current `all_easy_gym_life` project on June 21, 2026.

Purpose and scope

This document maps how the Flutter application, Node/Express backend, PostgreSQL schema, assets, platform runners, project configuration, and project documentation relate to one another.

The repository contains 147 Git-tracked files. Platform-generated files are grouped by responsibility where they have the same relationship to the Flutter application. Local-only or untracked files are listed separately because they are present in the working folder but are not part of the tracked repository.

Relationship labels used below:

- `loads`: reads a file at runtime
- `declares`: makes a dependency or asset available
- `calls`: sends a request to an API route
- `queries`: reads or writes database data
- `builds`: participates in producing a platform application
- `configures`: supplies build-time or runtime settings
- `documents`: explains or operationalizes another file or subsystem
- `planned`: described in documentation but not implemented in current code

1. Whole-system model

```
User
|
v
Flutter application
lib/main.dart
| loads frontend-safe configuration
| from assets/config/app.env
| or assets/config/app.env.example
|
| POST /api/auth/login
v
Node.js / Express backend
backend/src/server.js
| reads backend/.env
| verifies passwords with bcryptjs
| signs and verifies JWTs with jsonwebtoken
|
| parameterized SQL through pg
v
PostgreSQL / Aurora PostgreSQL
backend/sql/schema.sql
|
| app_user records
v
Role returned to Flutter
|
+--> business --> BusinessDashboard
|
+--> gim -----> GimUserDashboard

Planned extension:

Raspberry Pi vision worker
raspberry_pi/
|
| planned POST /api/vision/events
v
Express backend --> database summaries --> Flutter business dashboard
```

The backend is the security boundary. Flutter does not directly connect to the database, and backend secrets should never be placed in Flutter assets.

2. Runtime authentication sequence

```
1. main()
|
+--> AppConfigLoader.load()
|
| +--> assets/config/app.env, when present
| +--> assets/config/app.env.example, as fallback
|
+--> GimAccessApp
|
+--> LoginPage
|
+--> ApiAuthRepository.authenticate()
|
+--> POST {baseUrl}/api/auth/login
|
+--> backend/src/server.js
|
+--> SELECT app_user by email
+--> bcrypt.compare()
+--> jwt.sign()
|
+--> JSON: token + user + role
|
+--> AuthSession
|
+--> DashboardRouter
|
+--> BusinessDashboard
+--> GimUserDashboard
```

The login page also has temporary demo-entry callbacks. These create local `AuthSession` objects without calling Express:

```
Enter Business Demo Dashboard --> local business AppUser --> BusinessDashboard
Enter Member Demo Dashboard ---> local gim AppUser -----> GimUserDashboard
```

3. Flutter source model

The entire application-specific Dart implementation currently lives in `lib/main.dart` (3,981 lines). It has four logical layers.

3.1 Bootstrap and application state

```
main()
|
+--> AppConfigLoader
+--> GimAccessApp
    |
    +--> ApiAuthRepository
    +--> LoginAuditStore
    +--> AuthSession?
    +--> LoginPage when signed out
    +--> DashboardRouter when signed in
```

- `GimAccessApp` owns the active session and logout behavior.
- `LoginAuditStore` is in-memory only. It is not persisted to PostgreSQL.
- Demo sessions use placeholder tokens and do not authenticate with the server.

3.2 Authentication and configuration model

```
AppConfigLoader --> AppConfig --> ApiAuthRepository
    |               |               |
    |               |               +--> package:http
    |               +--> baseUrl
    +--> rootBundle
```

```
API JSON --> userRoleFromApi() --> UserRole
API JSON --> AppUser + token --> AuthSession
errors -----> AuthException
```

- `dart:convert` encodes login requests and decodes API responses.
- `package:http/http.dart` performs the login request.
- `package:flutter/services.dart` supplies `rootBundle` for configuration.
- Supported API roles are `business` and `gim`.
- The SQL schema also permits `developer_admin`, but Flutter currently rejects that role as unsupported.

3.3 Navigation and dashboard model

```
DashboardRouter
|
+--> UserRole.business
|
|   +--> BusinessDashboard
|   |
|   |   +--> _BusinessSidebar
|   |   +--> _BusinessHeaderCard
|   |   +--> _BusinessTbdCard
|   |   +--> _BusinessActionCard
|   |   +--> source filters
|   |   +--> _BusinessContentPanel
|   |   |
|   |   |   +--> Home
|   |   |   +--> Video Feed
|   |   |   +--> Data Filter
|   |   |   +--> Equipment Health
|   |   |   +--> Insights
|   |   |   +--> Gym Map
|   |
|   +--> UserRole.gim
|   |
|   |   +--> GimUserDashboard
|   |   |
|   |   |   +--> _MemberDashboardHeader
|   |   |   +--> _MemberWelcomeStrip
|   |   |   +--> responsive layout
|   |   |   |
|   |   |   |   +--> _MemberDashboardWide
|   |   |   |   +--> _MemberDashboardStacked
|   |   |   |   |
|   |   |   |   |   +--> user information
|   |   |   |   |   +--> calendar
|   |   |   |   |   +--> planning
|   |   |   |   |   +--> today's workout
|   |   |   |   |   +--> schedule
|   |   |   |   |   +--> workout log
|   |   |   |   |   +--> focus card
```

Business dashboard values, equipment states, map zones, video placeholders, and insights are currently representative UI data in `main.dart`. They are not yet loaded from backend analytics endpoints.

3.4 Shared Flutter UI helpers

```
Color system:      _EglPastels
Logo widget:       _EglLogoMark
Panel decoration:   _panelDecoration()
Member decoration:  _memberPanelDecoration()
Reusable business UI: status badges, source chips, stat tiles, map markers
Reusable member UI: pills, action cards, calendar tiles, chart painter
Legacy/shared cards: _DashboardFrame, _ProfileCard, _ServerCard,
                     _ConfigPanel, _DemoAccountPanel
```

`_EglLogoMark` loads `Project Pictures/EGL Company Logo.png`.

4. Backend and database model

```
Root package.json
|
+--> npm workspace: backend/
|
+--> backend/package.json
|
+--> start/dev --> backend/src/server.js
+--> seed:demo --> backend/scripts/seed-demo-users.js

backend/.env
|
+--> server.js
|
|   +--> Express middleware
|   |   +--> cors
|   |   +--> morgan
|   |   +--> express.json
|   |
|   +--> pg.Pool --> PostgreSQL
|   +--> bcryptjs --> password verification
|   +--> jsonwebtoken --> JWT creation and validation
|
+--> seed-demo-users.js
|
+--> pg.Client --> PostgreSQL
+--> bcryptjs --> password hashing
```

4.1 Implemented routes

Route	Implemented in	Database relationship	Flutter relationship
GET /	backend/src/server.js	None	Backend status page only
GET /api/health	backend/src/server.js	Executes SELECT 1	Not currently called by Flutter
POST /api/auth/login	backend/src/server.js	Reads app_user	Called by ApiAuthRepository
GET /api/auth/me	backend/src/server.js	Reads app_user by JWT subject	Not currently called by Flutter
POST /api/vision/events	Documentation only	Planned event storage	Planned dashboard input

4.2 Current database schema

```
app_user
|
+-- id          BIGINT identity primary key
+-- email       unique login identifier
+-- password_hash  bcrypt password hash
+-- role        business | gim | developer_admin
+-- display_name frontend display value
+-- is_active    account access switch
+-- created_at   creation timestamp
+-- updated_at   update timestamp
|
+-- index: idx_app_user_role
+-- index: idx_app_user_active
```

backend/scripts/seed-demo-users.js inserts or updates four demo users and depends on the app_user table created by backend/sql/schema.sql.

4.3 Planned data model

The local-only TECHNICAL_KNOWLEDGE.md and tracked project guides describe a future expansion beyond the implemented app_user table:

```
organizations 1 --> many locations
locations 1 -----> many equipment_assets
equipment_categories 1 --> many equipment_assets
equipment_assets 1 -----> many equipment_usage_logs
```

```

equipment_assets 1 -----> many maintenance events
locations 1 -----> many vision_events
users 1 -----> many login_audit records
aggregates -----> dashboard_metrics

```

These entities are design direction only. They are not present in the current tracked SQL schema.

5. Configuration and dependency relationships

5.1 Flutter dependency chain

```

pubspec.yaml
|
+--> Flutter SDK
+--> cupertino_icons
+--> http
+--> flutter_test
+--> flutter_lints
+--> declared Flutter assets
|
+--> resolved versions in pubspec.lock

analysis_options.yaml
|
+--> package:flutter_lints/flutter.yaml

.metadata
|
+--> Flutter project/tool migration metadata

```

5.2 Node dependency chain

```

package.json
|
+--> npm workspace: backend
+--> root scripts for backend and Flutter launch commands
+--> concurrently development dependency
|
+--> package-lock.json

backend/package.json
|
+--> express
+--> cors
+--> morgan
+--> dotenv
+--> pg
+--> bcryptjs
+--> jsonwebtoken
+--> nodemon
|
+--> resolved by root package-lock.json workspace data

```

5.3 Environment-file boundary

Frontend-safe:	
assets/config/app.env	local, ignored by Git
assets/config/app.env.example	tracked template and runtime fallback
Backend-secret:	
backend/.env	local, ignored by Git
backend/.env.example	tracked template

Important consistency note: the frontend template defaults to JWT issuer `egl-auth-service`, while `backend/.env.example` currently says `gim-auth-service`. The running backend and frontend should use the same issuer when token-based profile requests are added.

6. Asset relationships

Asset family	Declared or referenced by	Current status
Project Pictures/EGL Company Logo.png	pubspec.yaml, lib/main.dart	Active Flutter runtime asset
assets/config/app.env	pubspec.yaml, AppConfigLoader	Active when local file exists; ignored by Git
assets/config/app.env.example	pubspec.yaml, AppConfigLoader	Active fallback and tracked template
Project Pictures/Updated EGL Company Logo.png	No code reference	Tracked design alternative
assets/egl_logo.png	No pubspec.yaml declaration or code reference	Tracked but inactive
assets/gym_map.png	No pubspec.yaml declaration or code reference	Tracked but inactive
web/favicon.png	web/index.html	Active web browser icon
web/icons/*.png	web/index.html, web/manifest.json	Active web/PWA icons
Android mipmap-* icons	Android manifest/resource system	Active Android launcher icons
iOS app and launch images	iOS asset catalogs/project	Active iOS packaging assets
macOS app icons	macOS asset catalog/project	Active macOS packaging assets
Windows app_icon.ico	Runner.rc	Active Windows executable icon

7. Platform runner relationships

All platform folders are adapters around the same Dart entry point and Flutter asset bundle. They do not implement separate application business logic.

```
lib/main.dart + pubspec.yaml
|
+--> web/
|   +--> index.html starts flutter_bootstrap.js
|   +--> manifest.json defines installable web metadata
|   +--> favicon and PWA icons
|
+--> android/
|   +--> Gradle configuration
|   +--> AndroidManifest.xml
|   +--> MainActivity.kt --> FlutterActivity
|   +--> launch themes and launcher icons
|
+--> ios/
|   +--> Xcode project/workspace
|   +--> AppDelegate.swift and SceneDelegate.swift
|   +--> Info.plist and storyboards
|   +--> app/launch asset catalogs
|
+--> macos/
|   +--> Xcode project/workspace
|   +--> AppDelegate.swift and MainFlutterWindow.swift
|   +--> xcconfig, plist, entitlements, menu, icons
|
+--> linux/
|   +--> CMake build graph
|   +--> GTK runner
|   +--> generated plugin registrant files
|
+--> windows/
|   +--> CMake build graph
|   +--> Win32 runner
|   +--> generated plugin registrant files
```

```
+--> resources and executable manifest
```

7.1 Web file group

```
web/index.html
+--> web/favicon.png
+--> web/icons/Icon-192.png
+--> web/manifest.json
+--> Icon-192.png
+--> Icon-512.png
+--> Icon-maskable-192.png
+--> Icon-maskable-512.png
```

The web title and description still use the generated name `gym_map_demo`.

7.2 Android file group

```
android/settings.gradle.kts
+--> locates Flutter SDK
+--> includes app module

android/build.gradle.kts
android/gradle.properties
android/gradle/wrapper/gradle-wrapper.properties
+--> configure Gradle build environment

android/app/build.gradle.kts
+--> Flutter source at repository root
+--> applicationId com.example.gym_map_demo
+--> version values from Flutter

android/app/src/main/AndroidManifest.xml
+--> MainActivity.kt
+--> launcher icons
+--> launch/normal themes

debug/profile manifests
+--> platform-specific development overlays
```

7.3 iOS file group

```
ios/Flutter/*.xcconfig
+--> Flutter build settings

ios/Runner.xcodeproj + ios/Runner.xcworkspace
+--> Xcode build graph and schemes
+--> Runner source, plist, storyboards, and assets

AppDelegate.swift + SceneDelegate.swift
+--> Flutter engine/application lifecycle

Info.plist
+--> app identity, version variables, scenes, orientations

RunnerTests.swift
+--> native iOS test target placeholder
```


7.4 macOS file group

```
macos/Flutter/*.xcconfig
+--> Flutter build settings

GeneratedPluginRegistrant.swift
+--> generated plugin registration

Runner.xcodeproj + Runner.xcworkspace
+--> Xcode build graph and scheme

Runner/Configs/*.xcconfig
+--> product identity, modes, and compiler warnings

AppDelegate.swift + MainFlutterWindow.swift
+--> native lifecycle and Flutter window

Info.plist + entitlements
+--> app metadata and sandbox capabilities

RunnerTests.swift
+--> native macOS test target placeholder
```

7.5 Linux file group

```
linux/CMakeLists.txt
+--> linux/flutter/CMakeLists.txt
+--> linux/runner/CMakeLists.txt
+--> generated_plugins.cmake

linux/runner/main.cc
+--> my_application.cc/.h
+--> GTK window hosting Flutter

generated_plugin_registrant.cc/.h
+--> generated native plugin registration
```

7.6 Windows file group

```
windows/CMakeLists.txt
+--> windows/flutter/CMakeLists.txt
+--> windows/runner/CMakeLists.txt
+--> generated_plugins.cmake

windows/runner/main.cpp
+--> flutter_window.cpp/.h
+--> win32_window.cpp/.h
+--> utils.cpp/.h

Runner.rc
+--> resource.h
+--> resources/app_icon.ico

runner.exe.manifest
+--> Windows DPI/runtime metadata

generated_plugin_registrant.cc/.h
+--> generated native plugin registration
```

8. Documentation relationship tree

```
README.md
|
+--> directs newcomers to FIRST_READ_THIS.md
|
+--> FIRST_READ_THIS.md
|   +--> explains complete architecture and installation
|   +--> points developers to PROJECT_GUIDE.md
|   +--> points backend work to backend/BACKEND_GUIDE.md
|
+--> STAKEHOLDER_OVERVIEW.md
|   +--> non-technical product and status framing
|
+--> PROJECT_GUIDE.md
|   +--> frontend/backend setup
|   +--> environment templates
|   +--> local ports and emulator host rules
|
+--> DEMO_RUNBOOK.md
|   +--> startup sequence
|   +--> demo credentials
|   +--> presentation claims and fallback steps
|
+--> backend/BACKEND_GUIDE.md
|   +--> backend setup and implemented routes
|   +--> planned vision-event contract
|
+--> raspberry_pi/README.md
|   +--> planned edge worker role
|   +--> planned event payload
|   +--> planned POST /api/vision/events relationship
```

TECHNICAL_KNOWLEDGE.md is present locally and expands the planned relational model, but .gitignore excludes it from Git. It is therefore a local design reference rather than a shared tracked project document.

9. Complete tracked-file inventory

The following tree includes every tracked file, with repetitive platform image families compacted using counts and filename patterns.

```
all_easy_gym_life/
|
+-- .gitignore
+-- .metadata
+-- README.md
+-- FIRST_READ_THIS.md
+-- PROJECT_GUIDE.md
+-- DEMO_RUNBOOK.md
+-- STAKEHOLDER_OVERVIEW.md
+-- analysis_options.yaml
+-- pubspec.yaml
+-- pubspec.lock
+-- package.json
+-- package-lock.json
|
+-- lib/
|   +-- main.dart
|
+-- assets/
|   +-- config/
|   |   +-- app.env.example
|   +-- egl_logo.png
|   +-- gym_map.png
|
+-- Project Pictures/
|   +-- EGL Company Logo.png
|   +-- Updated EGL Company Logo.png
|
+-- backend/
|   +-- .env.example
|   +-- BACKEND_GUIDE.md
|   +-- package.json
|   +-- src/server.js
|   +-- scripts/seed-demo-users.js
```

```

|   +-- sql/schema.sql
|
+-- raspberry_pi/
|   +-- README.md
|
+-- web/
|   +-- index.html
|   +-- manifest.json
|   +-- favicon.png
|   +-- icons/
|       +-- Icon-192.png
|       +-- Icon-512.png
|       +-- Icon-maskable-192.png
|       +-- Icon-maskable-512.png
|
+-- android/ (19 tracked files)
|   +-- .gitignore
|   +-- settings.gradle.kts
|   +-- build.gradle.kts
|   +-- gradle.properties
|   +-- gradle/wrapper/gradle-wrapper.properties
|   +-- app/build.gradle.kts
|   +-- app/src/debug/AndroidManifest.xml
|   +-- app/src/profile/AndroidManifest.xml
|   +-- app/src/main/AndroidManifest.xml
|   +-- app/src/main/kotlin/com/example/gym_map_demo/MainActivity.kt
|   +-- app/src/main/res/drawable/launch_background.xml
|   +-- app/src/main/res/drawable-v21/launch_background.xml
|   +-- app/src/main/res/values/styles.xml
|   +-- app/src/main/res/values-night/styles.xml
|   +-- app/src/main/res/mipmap-{mdpi,hdpi,xhdpi,xxhdpi}/
|       +-- ic_launcher.png (5 files)
|
+-- ios/ (40 tracked files)
|   +-- .gitignore
|   +-- Flutter/AppFrameworkInfo.plist
|   +-- Flutter/Debug.xcconfig
|   +-- Flutter/Release.xcconfig
|   +-- Runner.xcodeproj/project.pbxproj
|   +-- Runner.xcodeproj/xcschemedata/xcschemes/Runner.xcscheme
|   +-- Runner.xcodeproj/project.xcworkspace/contents.xcworkspacedata
|   +-- Runner.xcodeproj/project.xcworkspace/xcschemedata/
|       +-- IDEWorkspaceChecks.plist
|       +-- WorkspaceSettings.xcsettings
|   +-- Runner.xcworkspace/contents.xcworkspacedata
|   +-- Runner.xcworkspace/xcschemedata/
|       +-- IDEWorkspaceChecks.plist
|       +-- WorkspaceSettings.xcsettings
|   +-- Runner/AppDelegate.swift
|   +-- Runner/SceneDelegate.swift
|   +-- Runner/Info.plist
|   +-- Runner/Runner-Bridging-Header.h
|   +-- Runner/Base.lproj/Main.storyboard
|   +-- Runner/Base.lproj/LaunchScreen.storyboard
|   +-- Runner/Assets.xcassets/AppIcon.appiconset/
|       +-- Contents.json
|       +-- Icon-App-*.png (15 files)
|   +-- Runner/Assets.xcassets/LaunchImage.imageset/
|       +-- Contents.json
|       +-- README.md
|       +-- LaunchImage.png
|       +-- LaunchImage@2x.png
|       +-- LaunchImage@3x.png
|   +-- RunnerTests/RunnerTests.swift
|
+-- macos/ (28 tracked files)
|   +-- .gitignore
|   +-- Flutter/Flutter-Debug.xcconfig
|   +-- Flutter/Flutter-Release.xcconfig
|   +-- Flutter/GeneratedPluginRegistrant.swift
|   +-- Runner.xcodeproj/project.pbxproj
|   +-- Runner.xcodeproj/xcschemedata/xcschemes/Runner.xcscheme
|   +-- Runner.xcodeproj/project.xcworkspace/xcschemedata/
|       +-- IDEWorkspaceChecks.plist
|   +-- Runner.xcworkspace/contents.xcworkspacedata
|   +-- Runner.xcworkspace/xcschemedata/IDEWorkspaceChecks.plist
|   +-- Runner/AppDelegate.swift
|   +-- Runner/MainFlutterWindow.swift
|   +-- Runner/Info.plist
|   +-- Runner/DebugProfile.entitlements

```

```

| +-- Runner/Release.entitlements
| +-- Runner/Base.lproj/MainMenu.xib
| +-- Runner/Configs/
| | +-- AppInfo.xcconfig
| | +-- Debug.xcconfig
| | +-- Release.xcconfig
| | +-- Warnings.xcconfig
| +-- Runner/Assets.xcassets/AppIcon.appiconset/
| | +-- Contents.json
| | +-- app_icon_{16,32,64,128,256,512,1024}.png
| +-- RunnerTests/RunnerTests.swift
|
+-- linux/ (10 tracked files)
| +-- .gitignore
| +-- CMakeLists.txt
| +-- flutter/CMakeLists.txt
| +-- flutter/generated_plugin_registrant.cc
| +-- flutter/generated_plugin_registrant.h
| +-- flutter/generated_plugins.cmake
| +-- runner/CMakeLists.txt
| +-- runner/main.cc
| +-- runner/my_application.cc
| +-- runner/my_application.h
|
+-- windows/ (18 tracked files)
| +-- .gitignore
| +-- CMakeLists.txt
| +-- flutter/CMakeLists.txt
| +-- flutter/generated_plugin_registrant.cc
| +-- flutter/generated_plugin_registrant.h
| +-- flutter/generated_plugins.cmake
| +-- runner/CMakeLists.txt
| +-- runner/main.cpp
| +-- runner/flutter_window.cpp
| +-- runner/flutter_window.h
| +-- runner/win32_window.cpp
| +-- runner/win32_window.h
| +-- runner/utils.cpp
| +-- runner/utils.h
| +-- runner/Runner.rc
| +-- runner/resource.h
| +-- runner/runner.exe.manifest
| +-- runner/resources/app_icon.ico

```

10. Local-only and generated working files

These files are present in the current working folder but are not part of the tracked 147-file tree:

- assets/config/app.env: ignored local frontend configuration
- backend/.env: ignored backend credentials and secret configuration, if

created

- TECHNICAL_KNOWLEDGE.md: ignored local architecture reference
- test_opencode.txt: currently untracked
- .dart_tool/, build/, node_modules/, and similar generated directories:

ignored dependency/build output when present

11. Current gaps and relationship risks

1. POST /api/vision/events is documented but not implemented.
2. Dashboard analytics are representative constants in lib/main.dart, not backend responses.
3. GET /api/health and GET /api/auth/me are implemented but unused by the current Flutter code.

4. developer_admin is allowed by SQL but unsupported by Flutter role parsing.
5. assets/egl_logo.png, assets/gym_map.png, and the updated company logo are tracked but not wired into Flutter.
6. Web and native platform metadata still use generated gym_map_demo and com.example identifiers rather than final Easy Gym Life branding.
7. The frontend configuration exposes database name and user values in the UI model. They are not secrets in the template, but the frontend does not need database connection details to call the API.
8. There is no tracked Dart test directory; native runner test files are still template placeholders.
9. LoginAuditStore is session-memory UI state and is separate from the planned persistent login_audit database entity.

12. Recommended modular target tree

The current single-file Flutter implementation works for a prototype, but the natural future file relationship would be:

```
lib/  
+-- main.dart  
+-- app/  
|   +-- app.dart  
|   +-- router.dart  
+-- config/  
|   +-- app_config.dart  
+-- auth/  
|   +-- data/api_auth_repository.dart  
|   +-- models/app_user.dart  
|   +-- models/auth_session.dart  
|   +-- presentation/login_page.dart  
+-- business/  
|   +-- presentation/business_dashboard.dart  
|   +-- widgets/  
+-- member/  
|   +-- presentation/member_dashboard.dart  
|   +-- widgets/  
+-- shared/  
    +-- theme/  
    +-- widgets/  
  
backend/src/  
+-- server.js  
+-- config/  
+-- middleware/authenticate-token.js  
+-- routes/  
|   +-- auth.js  
|   +-- health.js  
|   +-- vision.js  
+-- services/  
+-- repositories/
```

This is a recommendation, not a description of the current repository.